

A. Manuals

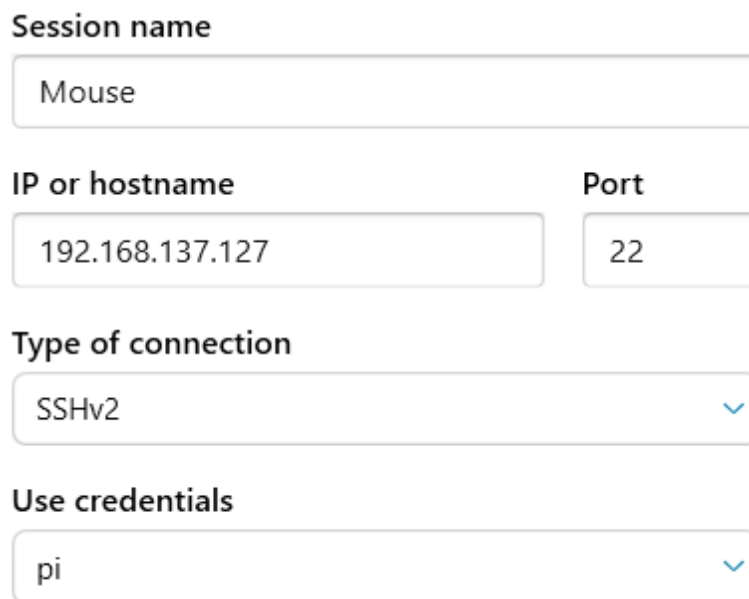
A.1. Connect to Nermo v4.1

1. Setup WiFi AP

A WiFi access point has to be configured with the SSID *LITTLELOSTROBOT* and the password *******. The Mouse will automatically connect to that WiFi.

2. Connect via SSH

Now a SSH connection to *pi@<IP>* can be established. For this log in procedure the same password as above can be used. Those - log in and password - can be saved as reusable credentials as in Figure A.1.



The image shows a configuration form for an SSH connection. It consists of several labeled input fields and dropdown menus. The labels are: 'Session name', 'IP or hostname', 'Port', 'Type of connection', and 'Use credentials'. The 'Session name' field contains the text 'Mouse'. The 'IP or hostname' field contains '192.168.137.127'. The 'Port' field contains '22'. The 'Type of connection' dropdown menu is set to 'SSHv2'. The 'Use credentials' dropdown menu is set to 'pi'. Each dropdown menu has a small blue downward arrow icon on its right side.

Session name	IP or hostname	Port	Type of connection	Use credentials
Mouse	192.168.137.127	22	SSHv2	pi

Figure A.1.: SSH Connection

A.2. Program Spine Board

1. Hardware Setup

To program the spine board three wires (GND, TX and RX) have to be soldered to the spine board (Figure A.2 left). After that step it can be connected via a USB to UART adapter to a computer (Figure A.2 right).

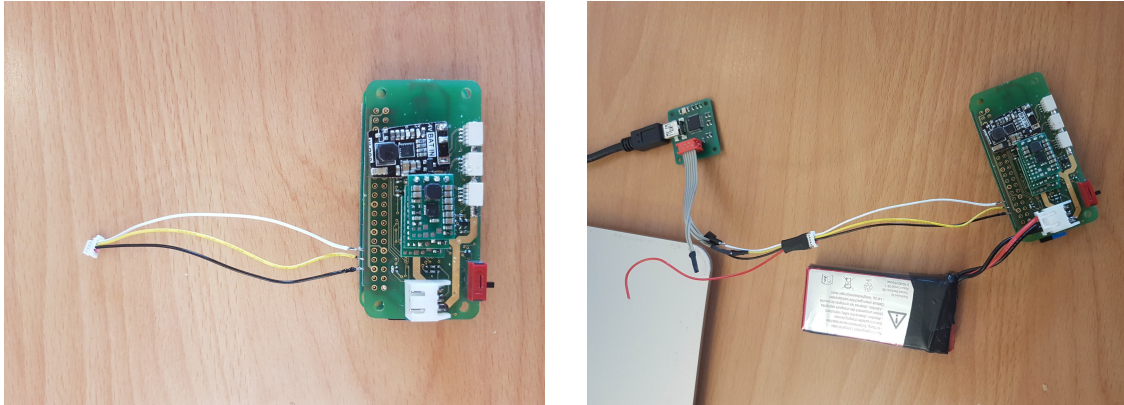


Figure A.2.: Wire and Complete Setup to Program Servo Board

2. SSH Connection

Now that the Hardware part is completed, the ssh connection (Figure A.3) to the Raspberry Pi has to be established as explained in section A.1.

```
⋮ Mouse x | +
Using username "pi".
Linux raspberrypi 4.14.98+ #1200 Tue Feb 12 20:11:02 GMT 2019 armv6l

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Mar 31 15:09:54 2020 from 192.168.137.1
pi@raspberrypi:~ $ minicom
```

Figure A.3.: Connection to Nermo v4.1

3. Minicom

With minicom started the available options can be printed with the command ?? (Figure A.4).

```
??
HBP Mouse Spine V0.8: Dec 13 2019, 15:37:45  DEBUG
Available Commands:
:ixxx      forward xxx to servo chain i [0..3]

!PS[+,-]   set power to servos
!PR[+,-][D] set power to RPI2W (D:delayed by 30sec)

!L[-,+,.]  set LED off/on/blinking
?S         send all sensor values (debug)

RESET      restart spine
BOOTL      enter boot-loader for reprogramming
PROG<C><S> reprogram servo <S>[0..4] on chain <C>[0..2,L,R,S]
```

Figure A.4.: Minicom Options

4. Disconnect Raspberry Pi

To disable the power-supply of the Raspberry Pi the command *!PRD* (Delayed to allow a proper shutdown of Raspberry Pi within the next 30sec) has to be executed. After that minicom can be closed with *Ctrl + A* and then *X*. Now the Raspberry Pi can be shut down with *sudo shutdown now*.

5. Serial Connection

A serial terminal has to be configured to connect to the port, where the USB to UART adapter is plugged in with the setting 1.000.000 Baud, parity none, 8 data bits and 1 stop bit (Figure A.5).

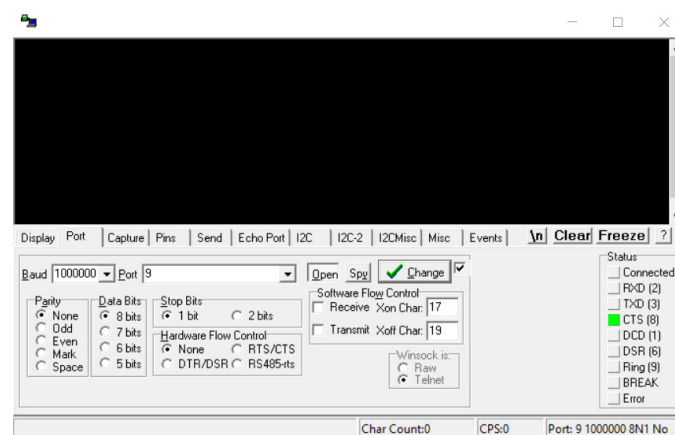


Figure A.5.: RealTerm Setup

6. Check Setup

To check the setup, the command `??` can be send as above (Set end of line to `CR` and `LF`). The output should be the same as in minicom before (Figure A.6).

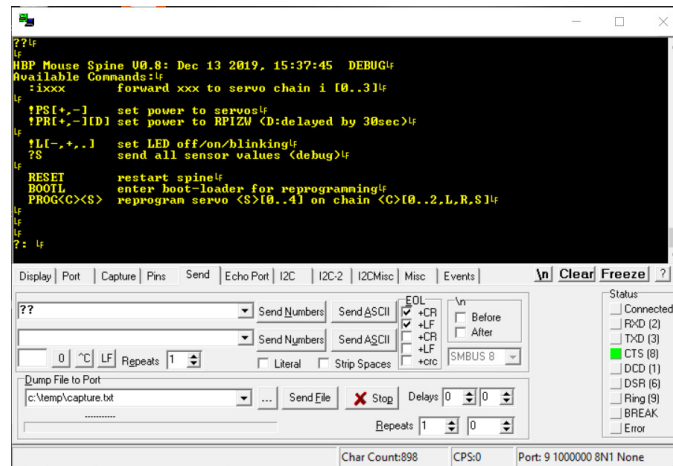


Figure A.6.: Send Options in RealTerm

7. Enter Boot Mode

Now as everything should be configured correctly, the command `BOOTL` allows to enter the boot mode on the spine board. If that was successful, the LED should stop blinking and the prompt as shown in Figure A.7 should appear.

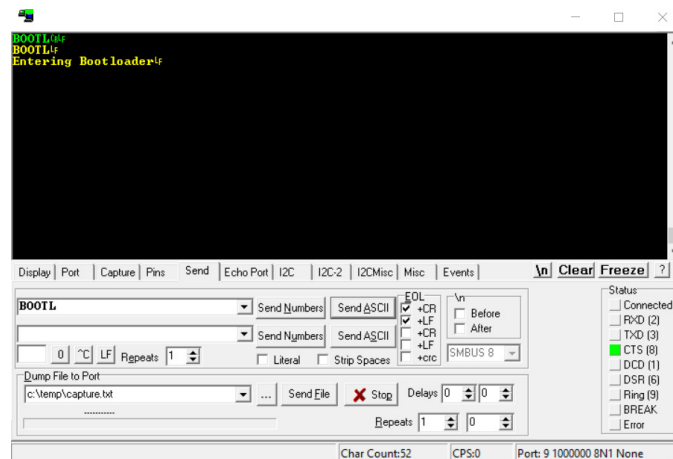


Figure A.7.: Enter Boot Mode

8. Close Port

After entering the boot mode the port in the serial terminal has to be closed again to allow the connection to the STM32CubeProgrammer (Figure A.8).

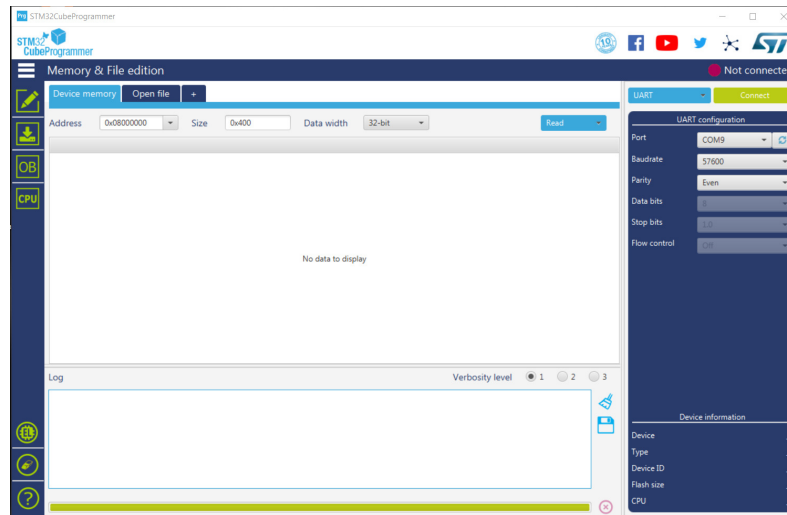


Figure A.8.: STM32CubeProgrammer

9. STM32CubeProgrammer

Now the connection to the STM32CubeProgrammer can be established (UART / 57600 Baud / parity even / 1 stop bit) as in Figure A.9.

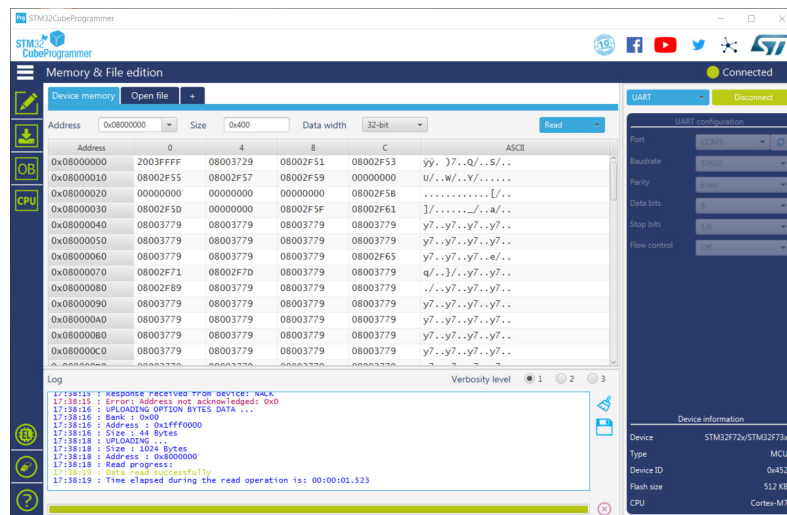


Figure A.9.: Connected

10. Programming

To program the spine board the compiled code can be uploaded to the spine board as *.elf* file via the menu *Erasing & Programming*.

A.3. Program Servo Board

The servo boards can be programmed in two ways. The first one is the preferred and easier one through the spine board via code forwarding. The second one is just needed if the servo is not connected to the spine board. If the servo board is completely new or crashed, the boot mode will have to be entered manually.

A.3.1. Through the Spine Board

The procedure to program the servo board through the spine board can be compared to programming the spine board itself. The only difference is that instead of sending *BOOTL* to the spine board, *PROG<ID>* is sent.

A.3.2. Without the Spine Board

To program the servo board without the spine board a special cable (Figure A.10) is needed to connect the servo board to the USB to UART adapter.

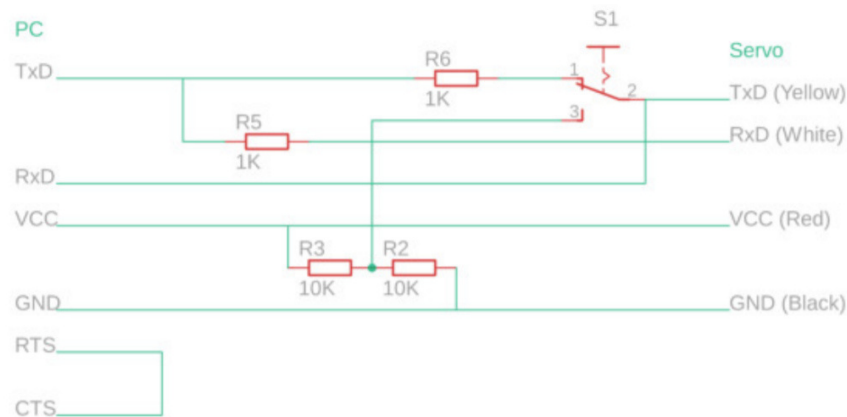


Figure A.10.: Cable to Connect Servo Board (Figure originally from: [27])

This cable has to be connected to the USB to UART adapter and the servo board. After this is done, the serial terminal (1.000.000 Baud, parity none, 8 data bits and 1 stop bit) can be connected to the port again. In this scenario the command to enter the boot mode is *<ID>BOOTL*, where *<ID>* is just the second digit of the servos ID (Set end of line to *CR* and *LF*). The next steps are the same as for programming the spine board.

A.3.3. Set Servo Board Manually to Boot Mode

In case the servo board is not programmed yet or crashed, the boot mode will have to be entered manually. Therefore the servo has to be disassembled because the pin *BOOT0* has to be set to VCC (3.3V) during the power up. This pin can be found on the backside of the servo board. The easiest procedure (recommended by Jörg Conradt) is probably to connect the two marked pins (Figure A.11) with tweezers and a second person reprogramming the board as usual.

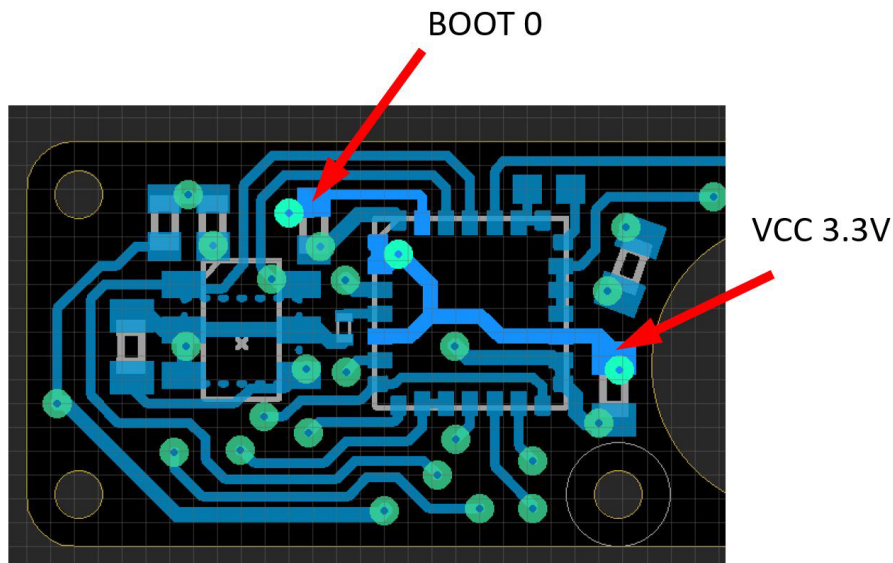


Figure A.11.: Position of the Pins (Figure originally from Jörg Conradt)

A.4. Set Servo Board ID

After soldering all components according to the schematics found in [27], the ID for each servo board has to be set according to the position it will be used for. The ID, which has to be set in hardware is the last digit of the servo board ID in Table A.1. Therefore, the two pads marked with *ID encoding* in Figure A.12 have to be connected as follows:

- ID 0: open (not connected)
- ID 1: solder bridge (0Ω)
- ID 2: resistor with $1k\Omega$
- ID 3: resistor with $10k\Omega$
- ID 4: resistor with $100k\Omega$

The suitable size for those resistors is 0402 (1mm * 0.5mm).

Table A.1.: Servo Board IDs

Position	ID	Position	ID	Position	ID
Left Hind Coil	00	Right Hind Coil	10	Spine Vertical	20
Left Hind Leg	01	Right Hind Leg	11	Spine Lateral	21
Left Front Coil	02	Right Front Coil	12	Spine Tail	22
Left Front Leg	03	Right Front Leg	13	Spine Neck	23
				Spine Head	24

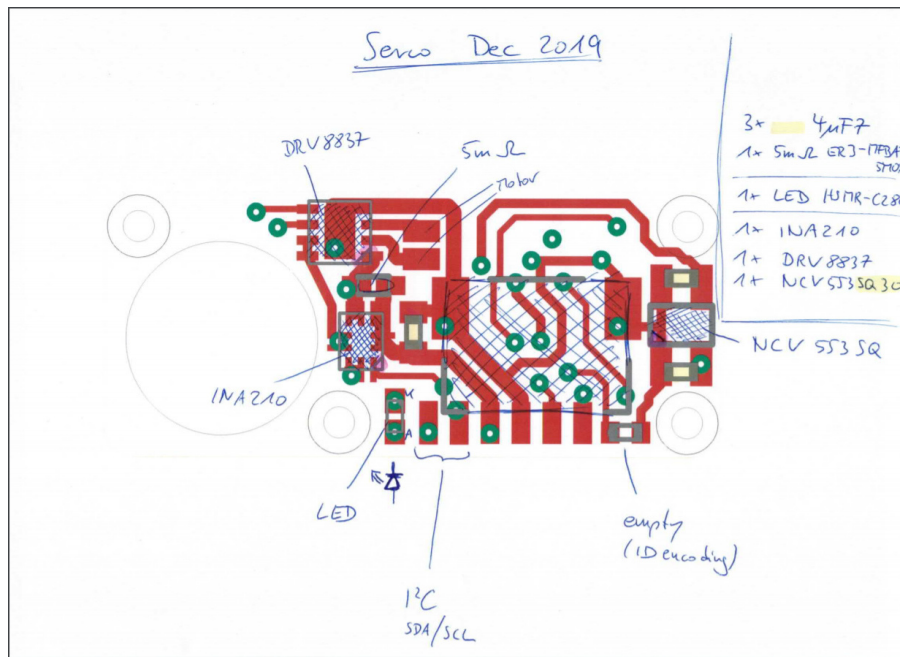


Figure A.12.: Set Servo Board ID (Figure originally from Jörg Conradt)

A.5. Wiring the Feet Setup

To wire the feet setup (see Figure A.13 for an overview) one should start with the connection of servo board and sensor board via four wires - SDA, SCL, V+, GND. After that procedure, the servo board can be attached to the servo via the blue and red cable - blue to *Motor 2* and red to *Motor 1* - and then fixed with screws. As a next step, the foot pressure sensor has to be connected to the sensor board via *VCC* respectively *VOUT* and *GND* (see [23] for more details). Therefor, the sensor has to be threaded through the designated slot in the foot first. Now, the sensor board can be screwed to the leg to finish the assembly.

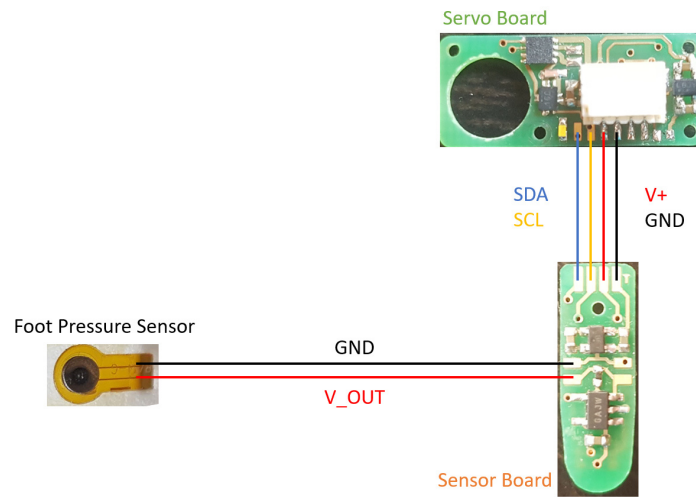


Figure A.13.: Soldering Tracks